

RAD-TOLERANT CLASS V, HIGH-SPEED PWM CONTROLLER

Check for Samples: [UC1825-SP](#)

FEATURES

- **QML-V Qualified**, SMD 5962-87681
- **Rad-Tolerant: 30 kRad (Si) TID** ⁽¹⁾
- **Compatible With Voltage- or Current-Mode Topologies**
- **Practical Operation Switching Frequencies to 1 MHz**
- **50-ns Propagation Delay-to-Output**
- **High-Current Dual Totem Pole Outputs (1.5 A Peak)**
- **Wide Bandwidth Error Amplifier**
- **Fully Latched Logic With Double-Pulse Suppression**
- **Pulse-by-Pulse Current Limiting**
- **Soft Start/Maximum Duty-Cycle Control**
- **Undervoltage Lockout With Hysteresis**
- **Low Start-Up Current (1.1 mA)**

(1) Radiation tolerance is a typical value based upon initial device qualification with dose rate = 10 mrad/sec. Radiation Lot Acceptance Testing is available - contact factory for details.

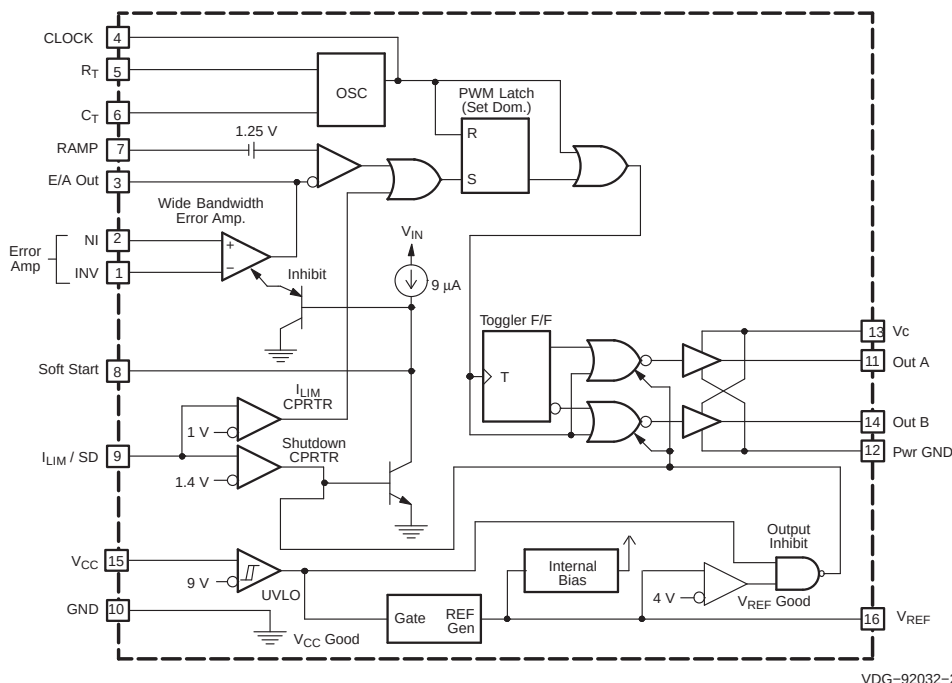
DESCRIPTION

The UC1825 PWM control device is optimized for high-frequency switched mode power supply applications. Particular care was given to minimizing propagation delays through the comparators and logic circuitry while maximizing bandwidth and slew rate of the error amplifier. This controller is designed for use in either current-mode or voltage mode systems with the capability for input voltage feed-forward.

Protection circuitry includes a current limit comparator with a 1-V threshold, a TTL compatible shutdown port, and a soft start pin which will double as a maximum duty-cycle clamp. The logic is fully latched to provide jitter-free operation and prohibit multiple pulses at an output. An undervoltage lockout section with 800 mV of hysteresis assures low start up current. During undervoltage lockout, the outputs are high impedance.

This device features totem pole outputs designed to source and sink high peak currents from capacitive loads, such as the gate of a power MOSFET. The on state is designed as a high level.

Figure 1. BLOCK DIAGRAM



VDG-92032-2



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

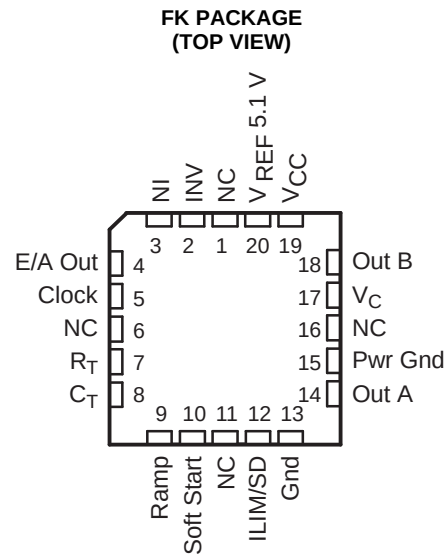
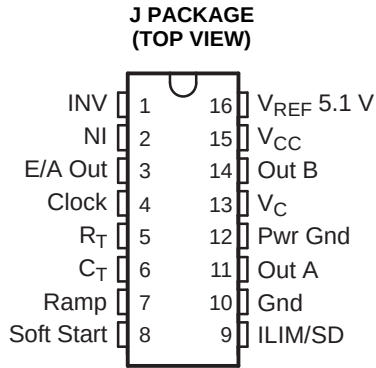
ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

Table 1. ORDERING INFORMATION⁽¹⁾

T_A	PACKAGE⁽²⁾	ORDERABLE PART NUMBER	TOP-SIDE MARKING
–55°C to 125°C	CDIP – J	5962-8768104VEA	UC1825J-SP
	LCCC – FK	5962-8768104V2A	UC1825FK-SP

(1) For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI web site at www.ti.com.

(2) Package drawings, thermal data, and symbolization are available at www.ti.com/packaging.



TERMINAL FUNCTIONS

NAME	NO.		I/O	DESCRIPTION
	J	FK		
Clock	4	5	O	Output of the internal oscillator
CT	6	8	I	Timing capacitor connection pin for oscillator frequency programming. The timing capacitor should be connected to the device ground using minimal trace length.
E/A Out	3	4	O	Output of the error amplifier for compensation
Gnd	10	13	-	Analog ground return pin
ILIM/SD	9	12	I	Input to the current limit comparator and the shutdown comparator
INV	1	2	I	Inverting input to the error amplifier
NC		1, 6, 11, 16	-	No connection
NI	2	3	I	Non-inverting input to the error amplifier
Out A	11	14	O	High-current totem pole output A of the on-chip drive stage
Out B	14	18	O	High-current totem pole output B of the on-chip drive stage
Pwr Gnd	12	15	-	Ground return pin for the output driver stage
Ramp	7	9	I	Non-inverting input to the PWM comparator with 1.25-V internal input offset. In voltage mode operation this serves as the input voltage feed-forward function by using the CT ramp. In peak current mode operation, this serves as the slope compensation input.
RT	5	7	I	Timing resistor connection pin for oscillator frequency programming
Soft Start	8	10	I	Soft-start input pin which also doubles as the maximum duty cycle clamp
VC	13	17	-	Power supply pin for the output stage. This pin should be bypassed with a 0.1-μF monolithic ceramic low ESL capacitor with minimal trace lengths.
VCC	15	19	-	Power supply pin for the device. This pin should be bypassed with a 0.1-μF monolithic ceramic low ESL capacitor with minimal trace lengths.
VREF5.1 V	16	20	O	5.1-V reference. For stability, the reference should be bypassed with a 0.1-μF monolithic ceramic low ESL capacitor and minimal trace length to the ground plane.

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

			UNIT
Supply voltage	V_C, V_{CC}	30	V
Output current, source or sink, Out A, Out B	DC	0.5	A
	Pulse (0.5 μ s)	2.0	
Analog inputs	INV, NI, Ramp	–0.3 to 7	V
	Soft Start, ILIM/SD	–0.3 to 6	
Clock output current	Clock	–5	mA
Error amplifier output current	E/A Out	5	
Soft-start sink current	Soft Start	20	
Oscillator charging current	R_T	–5	
Power dissipation		1	W
Storage temperature range		–65 to 150	°C
Lead temperature (soldering, 10 seconds)		300	

(1) All voltages are with respect to GND; all currents are positive into, negative out of part; pin numbers refer to DIL-16 package.

RECOMMENDED OPERATING CONDITIONS

over operating free-air temperature range ($T_A = T_J = -55^\circ\text{C}$ to 125°C), unless otherwise noted.

	MIN	MAX	UNIT
V_{CC} Supply voltage	10	30	V
Sink/source output current (continuous or time average)	0	100	mA
Reference load current	0	10	mA

THERMAL RATINGS TABLE

PACKAGE	θ_{JA} (°C/W)	θ_{JC} (°C/W)
DIL-16 (J)	80–120	28 ⁽¹⁾
LCC-20 (FK)	70–80	20 ⁽¹⁾

(1) θ_{JC} data values stated were derived from MIL-STD-1835B. MIL-STD-1835B states that the baseline values shown are worst case (mean + 2s) for a 60 × 60 mil microcircuit device silicon die and applicable for devices with die sizes up to 14400 square mils. For device die sizes greater than 14400 square mils use the following values; dual-in-line, 11°C/W; flat pack 10°C/W; pin grid array, 10°C/W.

ELECTRICAL CHARACTERISTICS

Unless otherwise stated, these specifications apply for $R_T = 3.65 \text{ k}\Omega$, $C_T = 1 \text{ nF}$, $V_{CC} = 15 \text{ V}$, $-55^\circ\text{C} < T_A < 125^\circ\text{C}$, $T_A = T_J$

PARAMETERS	TEST CONDITIONS	MIN	TYP	MAX	UNIT
REFERENCE					
Output voltage	$T_J = 25^\circ\text{C}$, $I_O = 1 \text{ mA}$	5.05	5.10	5.15	V
Line regulation	$10 \text{ V} < V_{CC} < 30 \text{ V}$		2	20	mV
Load regulation	$1 \text{ mA} < I_O < 10 \text{ mA}$		5	20	mV
Total output variation	Line, load, temperature	5.0		5.2	V
Output noise voltage	$10 \text{ Hz} < f < 10 \text{ kHz}$		50		μV
Short-circuit current	$V_{REF} = 0 \text{ V}$	-15	-50	-100	mA
OSCILLATOR SECTION					
Initial accuracy	$T_J = 25^\circ\text{C}$	360	400	440	kHz
Voltage stability	$10 \text{ V} < V_{CC} < 30 \text{ V}$		0.2%	2%	
Temperature stability	$T_{MIN} < T_A < T_{MAX}$		5%	16%	
Total variation	Line, Temperature	340		460	kHz
Clock out high		3.9	4.5		V
Clock out low			2.3	2.9	V
Ramp peak ⁽¹⁾		2.6	2.8	3.0	V
Ramp valley ⁽¹⁾		0.7	1.0	1.25	V
Ramp valley to peak ⁽¹⁾		1.6	1.8	2.1	V
ERROR AMPLIFIER					
Input offset voltage				10	mV
Input bias current			0.6	3	μA
Input offset current			0.1	1	μA
Open-loop gain	$1 \text{ V} < V_O < 4 \text{ V}$	60	95		dB
CMRR	$1.5 \text{ V} < V_{CM} < 5.5 \text{ V}$	75	95		dB
PSRR	$10 \text{ V} < V_{CC} < 30 \text{ V}$	85	110		dB
Output sink current	$V_{E/AOut} = 1 \text{ V}$	1	2.5		mA
Output source current	$V_{E/AOut} = 4 \text{ V}$	-0.5	-1.3		mA
Output high voltage	$I_{E/AOut} = -0.5 \text{ mA}$	4.0	4.7	5.0	V
Output low voltage	$I_{E/AOut} = 1 \text{ mA}$	0	0.5	1.0	V
Gain bandwidth product ⁽¹⁾	$f = 200 \text{ kHz}$	5	10.5		MHz
Slew rate ⁽¹⁾		4	9		V/ μs
PWM COMPARATOR					
Ramp bias current	$V_{Ramp} = 0 \text{ V}$		-1	-5	μA
Duty cycle range		0%		80%	
E/A out zero dc threshold	$V_{Ramp} = 0 \text{ V}$	1.1	1.25		V
Delay to output ⁽¹⁾			50	80	ns
SOFT-START					
Charge current	$V_{Soft Start} = 0.5 \text{ V}$	3	9	20	μA
Discharge current	$V_{Soft Start} = 1 \text{ V}$	1			mA
CURRENT LIMIT/SHUTDOWN					
Current limit/shutdown bias current	$0 < V_{ILIM/SD} < 4 \text{ V}$			15	μA
Current limit threshold		0.9	1.0	1.1	V
Shutdown threshold		1.25	1.40	1.55	V
Delay to output ⁽¹⁾			50	80	ns

(1) Parameters ensured by design and/or characterization, if not production tested.

ELECTRICAL CHARACTERISTICS (continued)

Unless otherwise stated, these specifications apply for $R_T = 3.65 \text{ k}\Omega$, $C_T = 1 \text{ nF}$, $V_{CC} = 15 \text{ V}$, $-55^\circ\text{C} < T_A < 125^\circ\text{C}$, $T_A = T_J$

PARAMETERS	TEST CONDITIONS	MIN	TYP	MAX	UNIT
OUTPUT					
Low-level output voltage	$I_{OUT} = 20 \text{ mA}$		0.25	0.40	V
	$I_{OUT} = 200 \text{ mA}$		1.2	2.2	V
High-level output voltage	$I_{OUT} = -20 \text{ mA}$	13.0	13.5		V
	$I_{OUT} = -200 \text{ mA}$	12.0	13.0		V
Collector leakage	$V_C = 30 \text{ V}$		10	500	μA
Rise/fall time ⁽²⁾	$C_L = 1 \text{ nF}$		30	75	ns
UNDER-VOLTAGE LOCKOUT					
Start threshold		8.8	9.2	9.6	V
UVLO hysteresis		0.4	0.8	1.2	V
SUPPLY CURRENT SECTION					
Startup current	$V_{CC} = 8 \text{ V}$		1.1	2.5	mA
I_{CC}	$V_{INV} = V_{Ramp} = V_{ILIM/SD} = 0 \text{ V}$, $V_{NI} = 1 \text{ V}$		22	33	mA

(2) Parameters ensured by design and/or characterization, if not production tested.

PRINTED CIRCUIT BOARD LAYOUT CONSIDERATIONS

High speed circuits demand careful attention to layout and component placement. To ensure proper performance of the UC1825 follow these rules:

1. Use a ground plane.
2. Damp or clamp parasitic inductive kick energy from the gate of driven MOSFETs. Do not allow the output pins to ring below ground. A series gate resistor or a shunt 1-A Schottky diode at the output pin serves this purpose.
3. Bypass V_{CC} , V_C , and V_{REF} . Use 0.1- μ F monolithic ceramic capacitors with low equivalent series inductance. Allow less than 1-cm of total lead length for each capacitor between the bypassed pin and the ground plane.
4. Treat the timing capacitor, C_T , like a bypass capacitor.

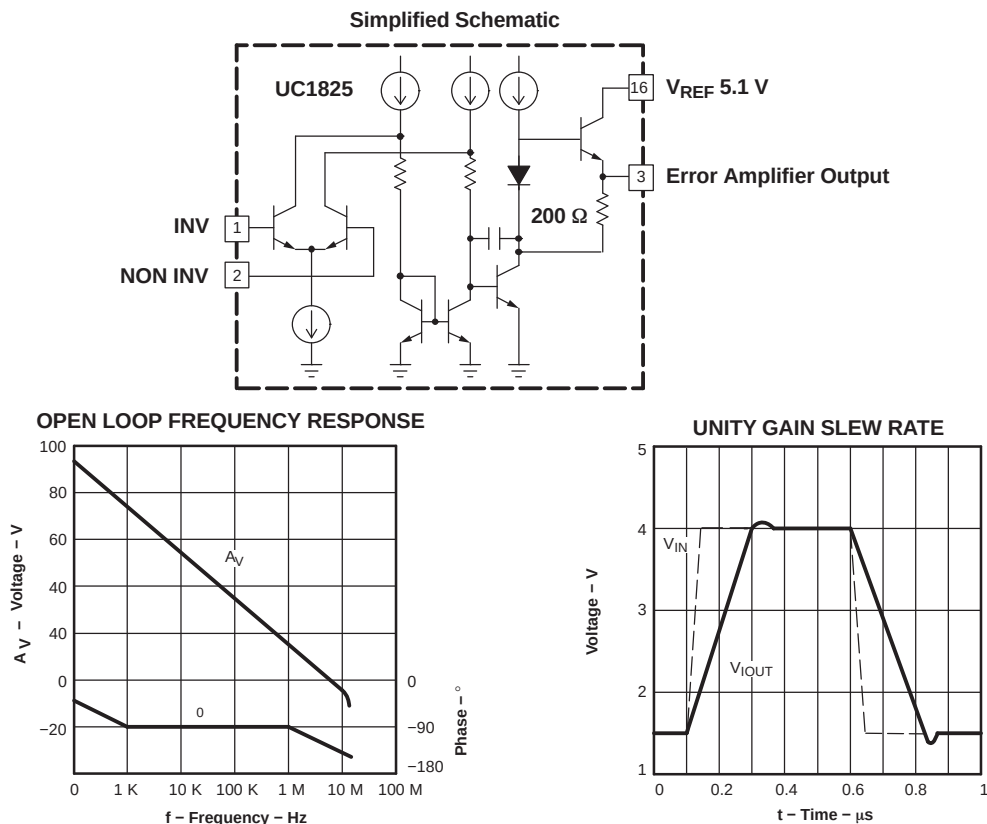


Figure 2. Error Amplifier

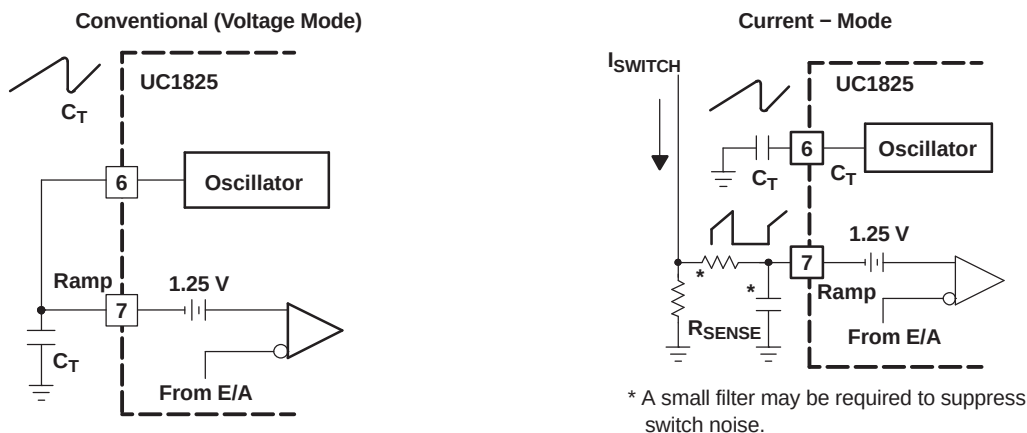


Figure 3. PWM Applications

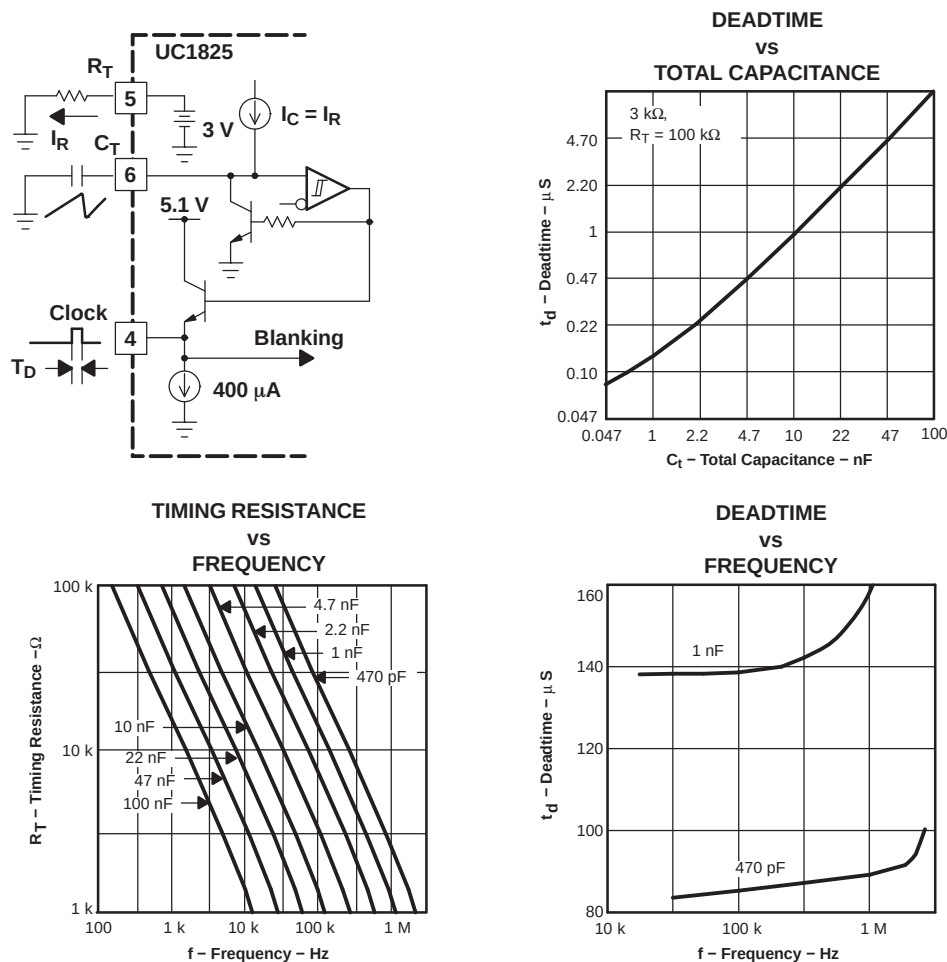


Figure 4. Oscillator Circuit

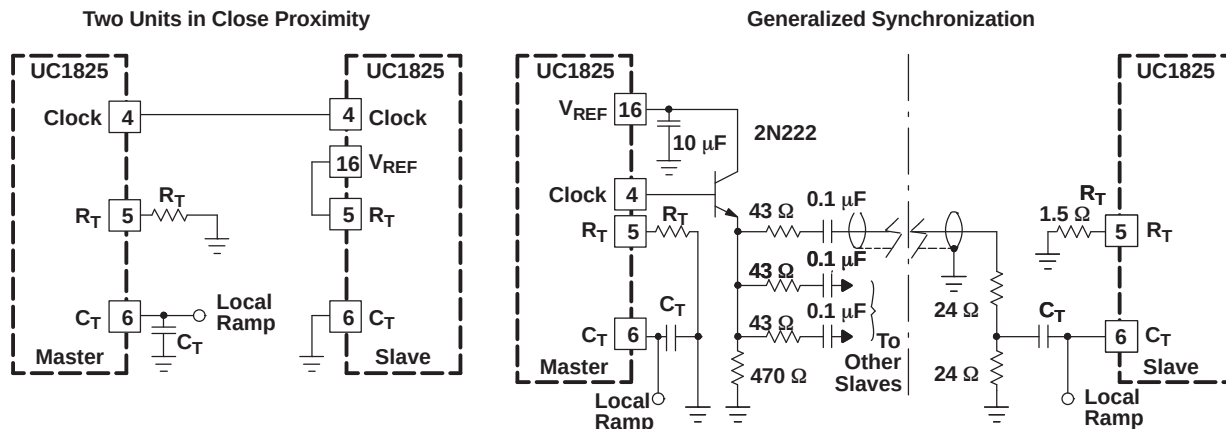


Figure 5. Synchronized Operation

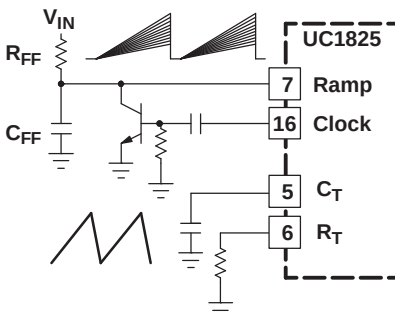


Figure 6. Forward Technique for Off-Line Voltage Mode Application

The circuit shown in Figure 6 will achieve a constant volt-second product clamp over varying input voltages. The ramp generator components, R_T and C_R are chosen so that the ramp at the ILIM/SD pin crosses the 1-V threshold at the same time the desired maximum volt-second product is reached. The delay through the functional nor block must be such that the ramp capacitor can be completely discharged during the minimum deadtime.

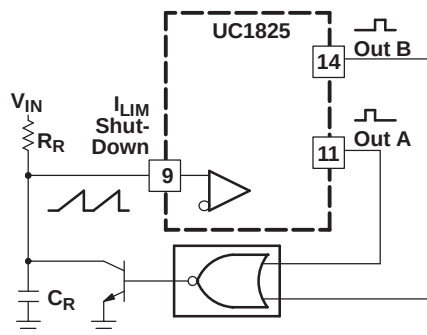


Figure 7. Constant Volt-Second Clamp Circuit

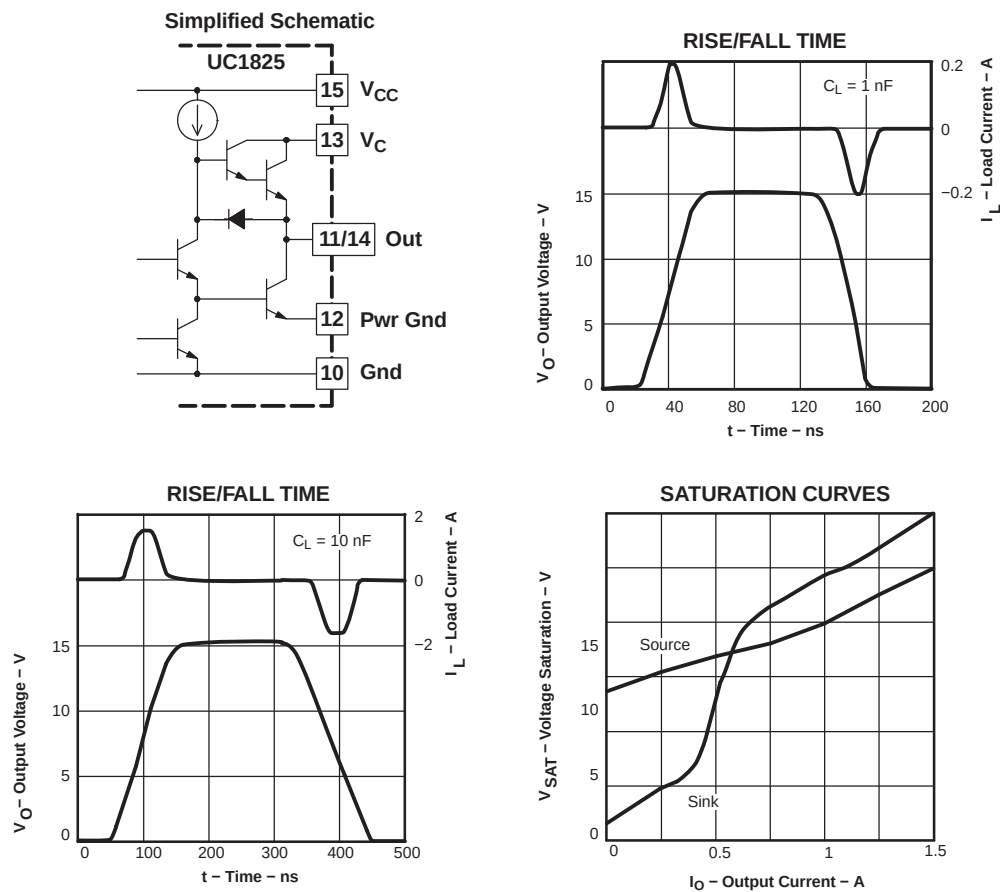


Figure 8. Output Section

The circuit in [Figure 8](#) is useful for exercising many of the UC1825 functions and measuring their specifications. As with any wideband circuit, careful grounding and bypass procedures should be followed. The use of a ground plane is highly recommended.

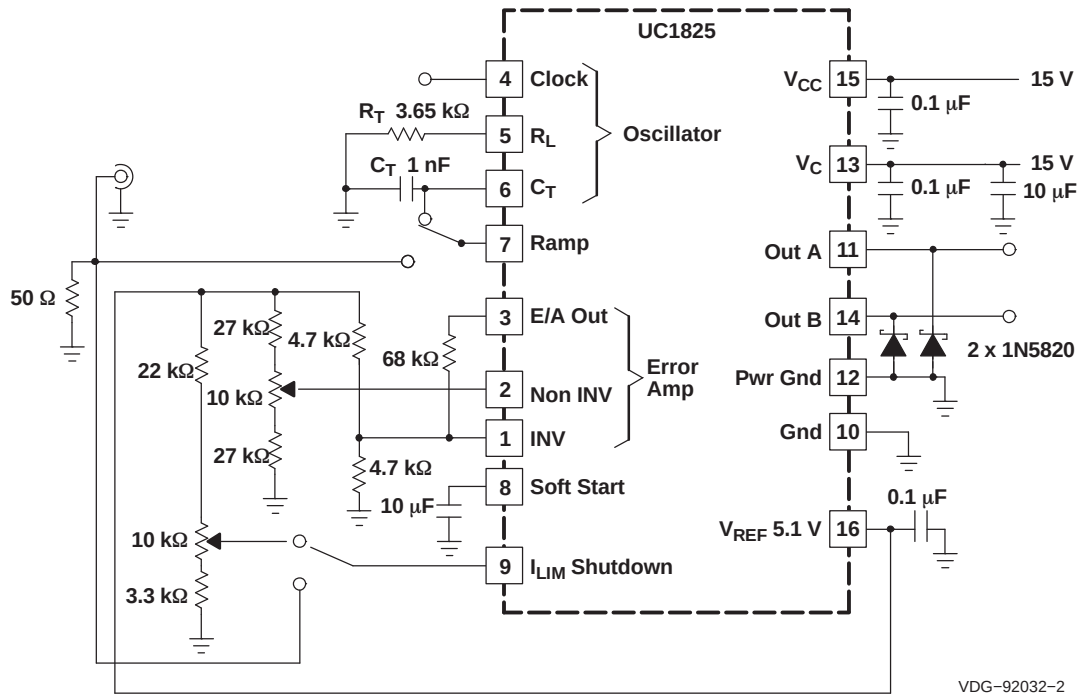
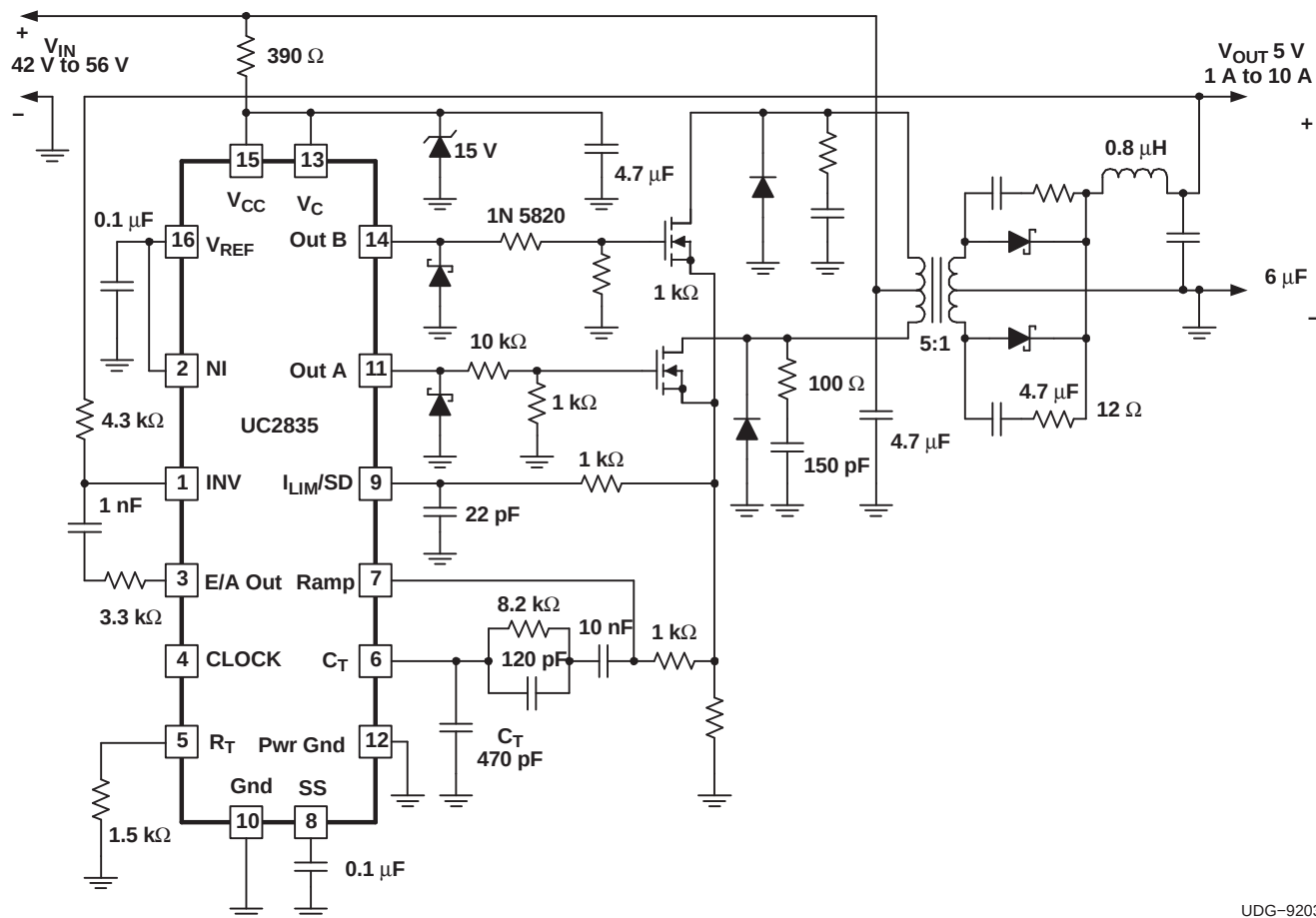


Figure 9. Open-Loop Laboratory Test Fixture



UDG-92033

Figure 10. Design Example: 50 W, 48-V to 5-V DC-to-DC Converter – 1.5-MHz Clock Frequency

REVISION HISTORY

Changes from Original (January, 2009) to Revision A

Page

-
- Added MAX spec of 16% to temperature stability parameter [5](#)
-

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-8768101V2A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 8768101V2A UC1825L QMLV
5962-8768101V2A.A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 8768101V2A UC1825L QMLV
5962-8768101VEA	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8768101VE A UC1825JQMLV
5962-8768101VEA.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8768101VE A UC1825JQMLV
5962-8768104V2A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 8768104V2A UC1825FK -SP
5962-8768104V2A.A	Active	Production	LCCC (FK) 20	55 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962- 8768104V2A UC1825FK -SP
5962-8768104VEA	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8768104VE A UC1825J-SP
5962-8768104VEA.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8768104VE A UC1825J-SP

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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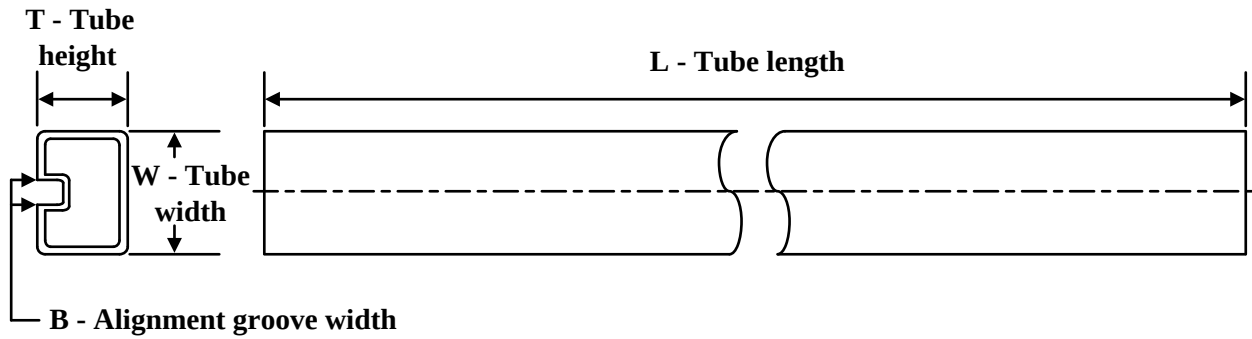
OTHER QUALIFIED VERSIONS OF UC1825-SP :

- Catalog : [UC1825](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
5962-8768101V2A	FK	LCCC	20	55	506.98	12.06	2030	NA
5962-8768101V2A.A	FK	LCCC	20	55	506.98	12.06	2030	NA
5962-8768104V2A	FK	LCCC	20	55	506.98	12.06	2030	NA
5962-8768104V2A.A	FK	LCCC	20	55	506.98	12.06	2030	NA

GENERIC PACKAGE VIEW

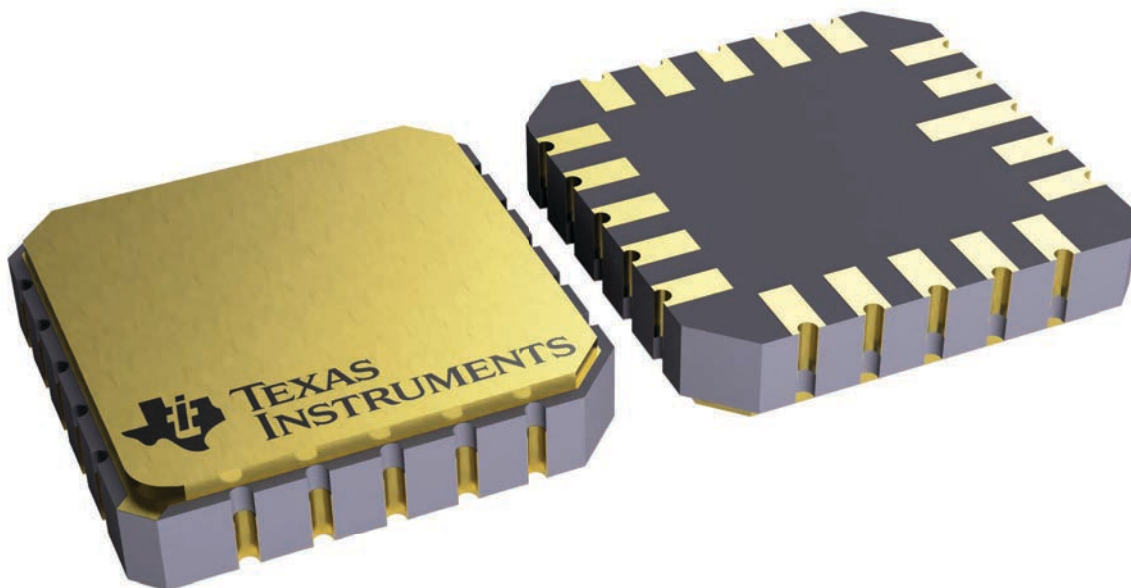
FK 20

LCCC - 2.03 mm max height

8.89 x 8.89, 1.27 mm pitch

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

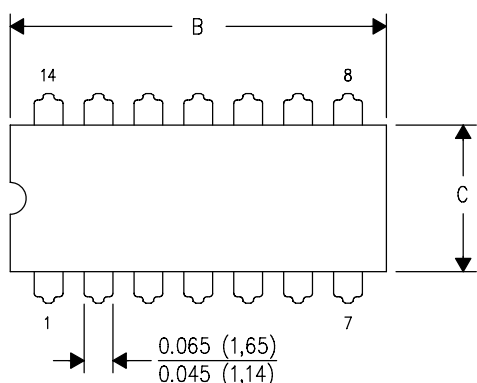


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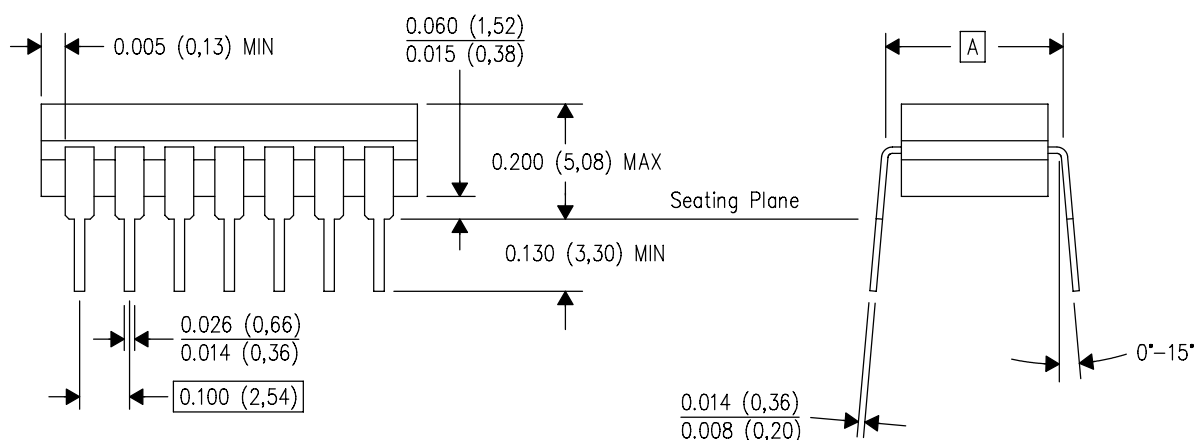
J (R-GDIP-T**)

14 LEADS SHOWN

CERAMIC DUAL IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)



4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. This package is hermetically sealed with a ceramic lid using glass frit.
 - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
 - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.

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